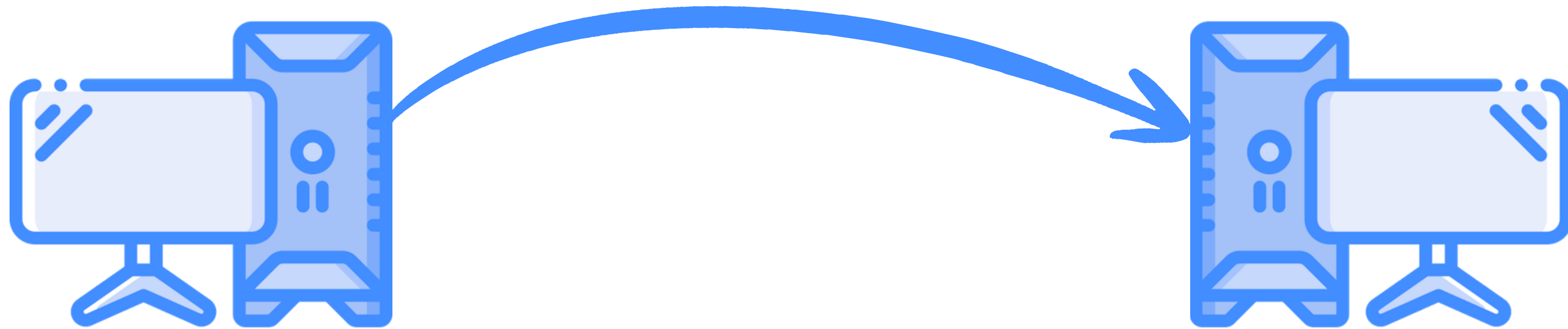


Networking and Communications



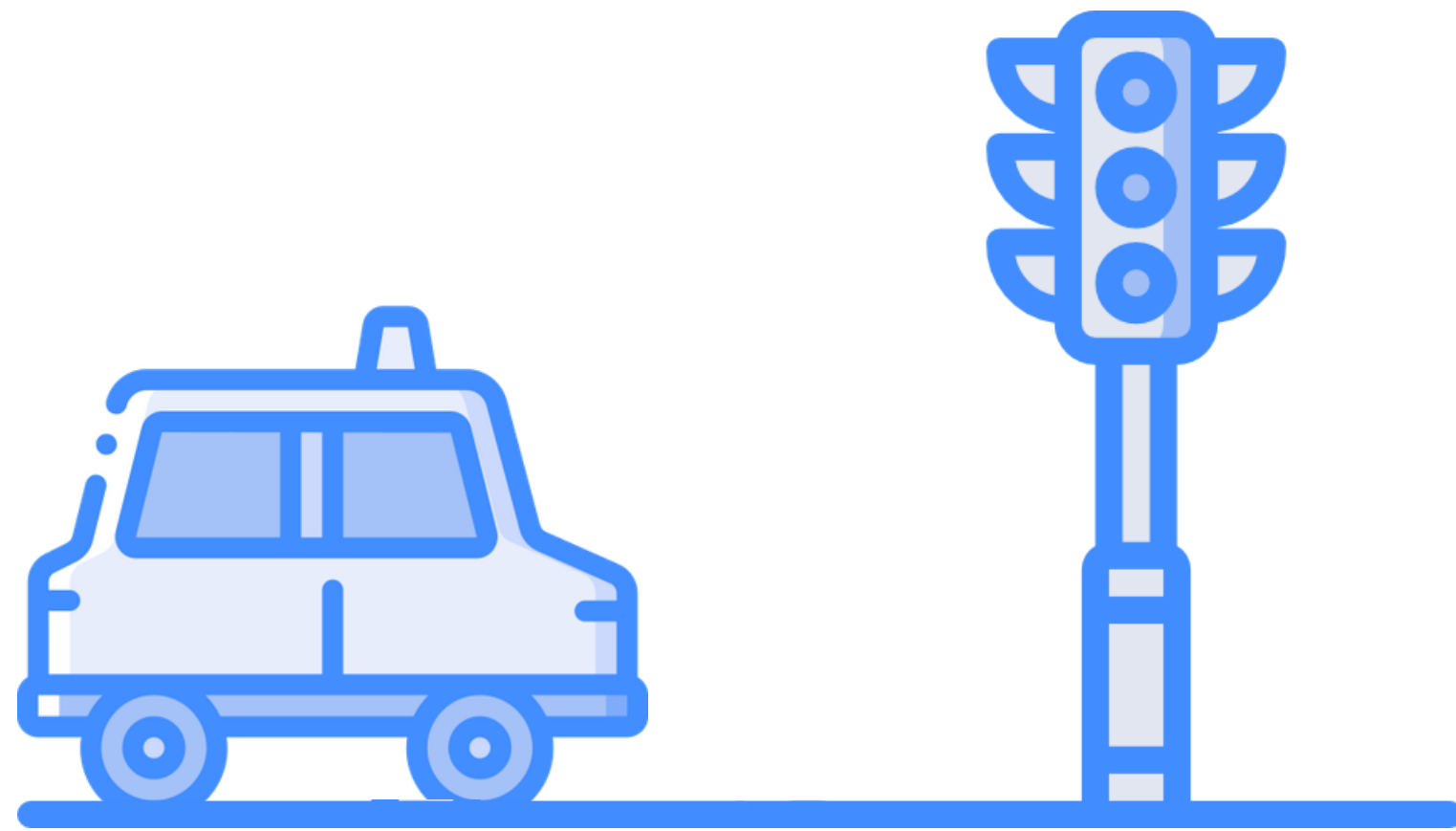
Networking Fundamentals

What is a Network



Basically, a network means that two or more connected computers are sharing resources with each another.

Protocols (Traffic Light Analogy)



Imagine that you are driving a car. The traffic lights, signs, and road rules are the protocols that prevent chaos and ensure everyone gets to their destination safely.

Green Light ● - GO

Yellow Light ● - SLOWDOWN

Red Light ● - STOP

What are protocols in Networking?



Similar to the simplified explanation regarding the traffic light analogy, protocols are rules governing how machines exchange data and enable effective communication. They are essential for maintaining order

Web Browsing ● - 80 (HTTP), 443 (HTTPS)

Email Services ● - 143 (IMAP), 25 (SMTP)

File Transfers ● - 21 (FTP), 22 (SFTP)

Connection Types

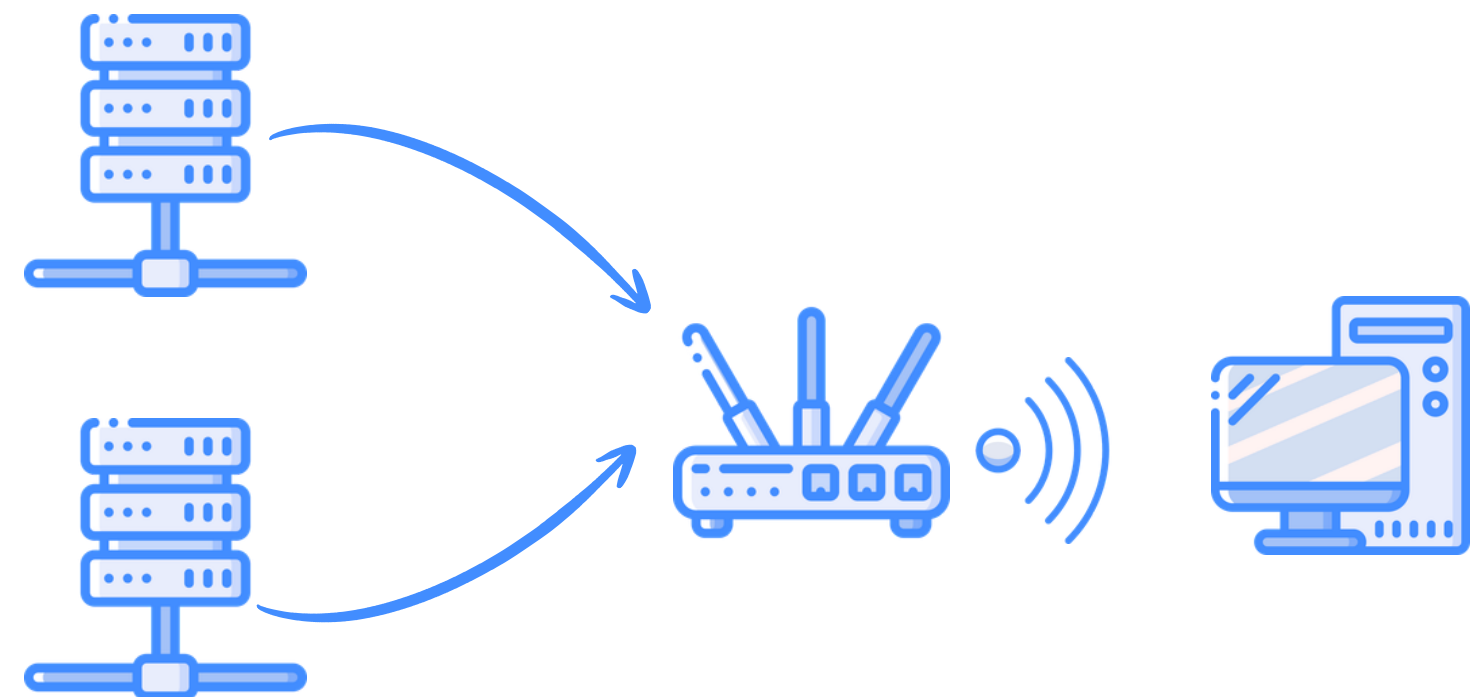
Peer-to-peer

- **All computers on the network are peers**
 - **No Dedicated servers**
 - **There's no centralized control over shared resources**
- **Any Devices can share resources as it pleases**
- **All computers can act as either a client or a server**
- **Easy to setup, and common in homes and small businesses**

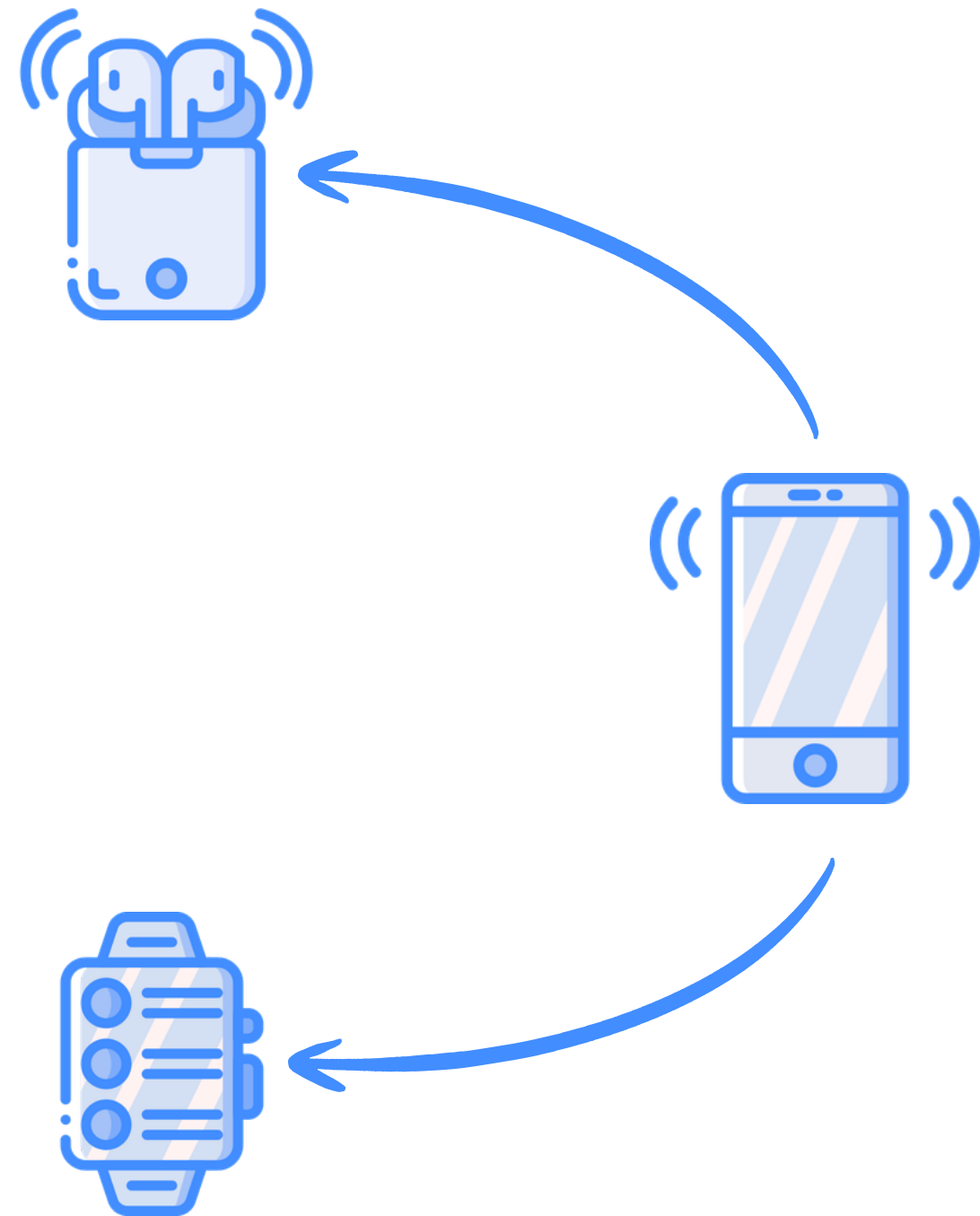


Client-Server

- **All network is composed of client and servers**
 - **Server provides the resources**
 - **Client receives the resources**
- **Servers provide centralized control**
- **Centralizes user accounts, security and access controls to simplify administration**
- **More Difficult to setup and requires experience**



Personal Area Network



(PAN)

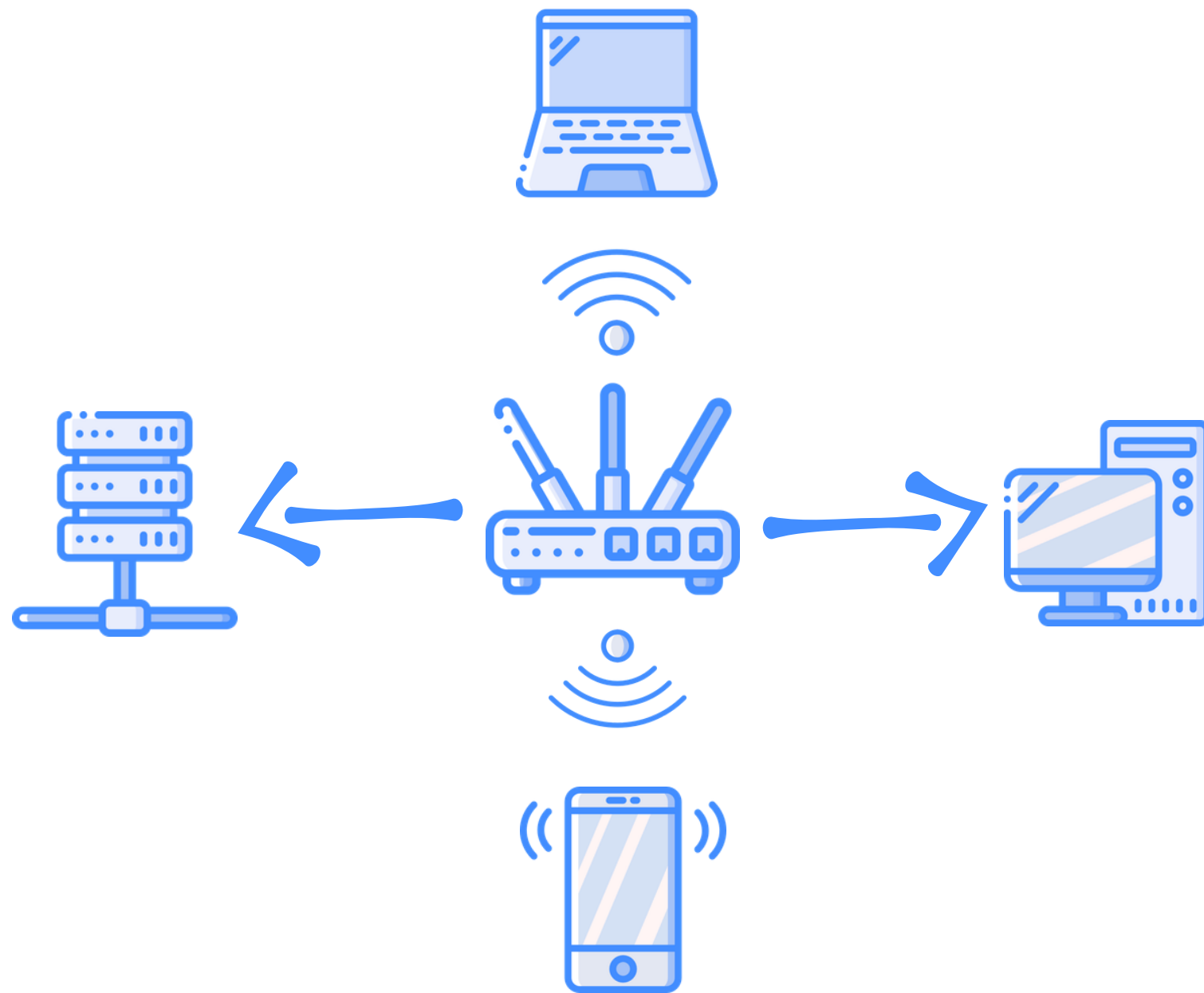
- **Ultra Small Networks used for personal use to share data from one device to another**

Examples

- **Smart Phone to Laptop**
- **Smart Watch to Smart Phone**
- **Smart Phone Hands-Free Car Calling**
- **Heart Rate Monitor to Smart Phone**

Local Area Network

(LAN)

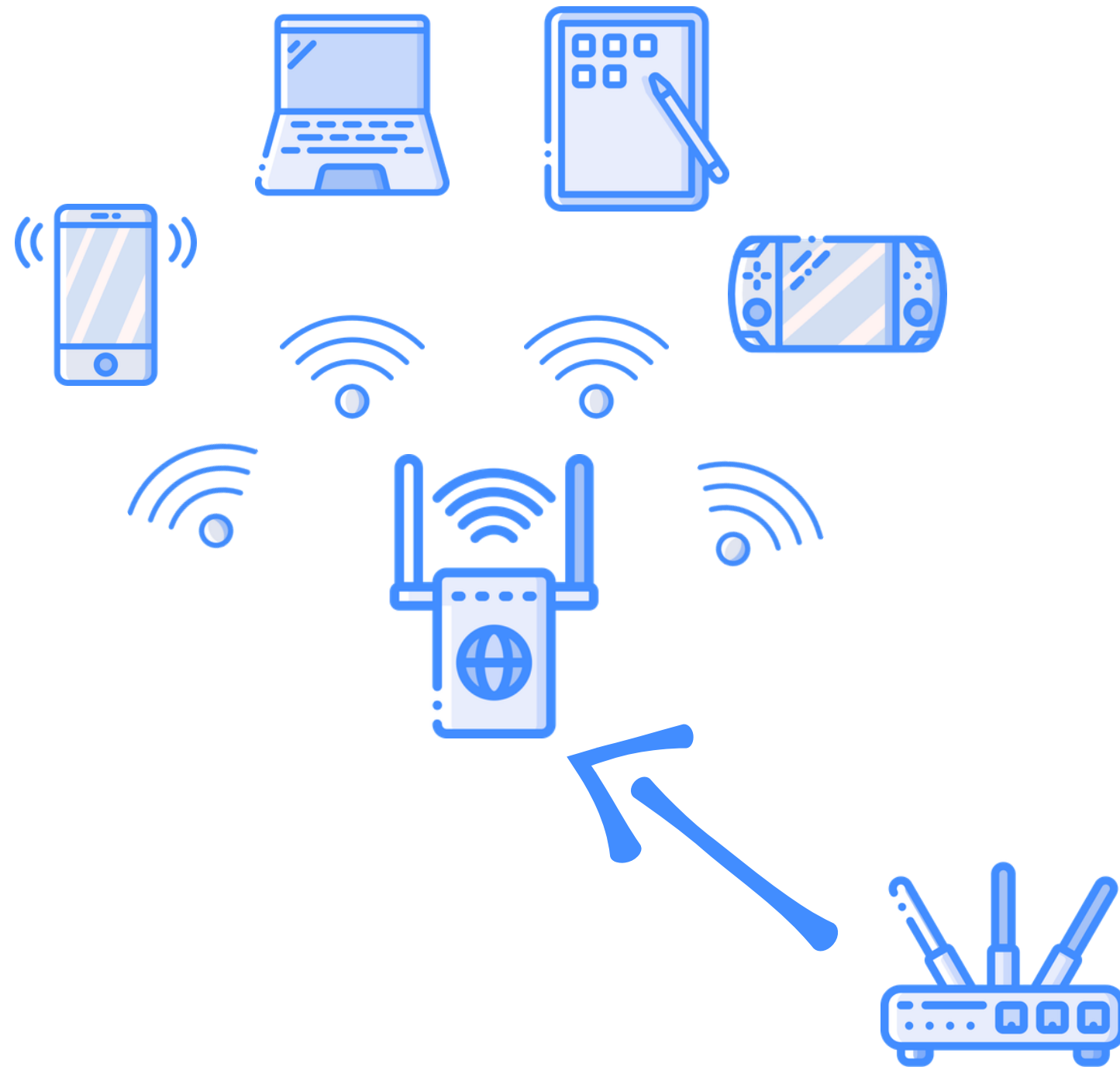


- **A computer network within a small geographical area, such as a single room, building or group of buildings.**
- **Considered to be self-contained:**
 - **All devices are directly connected via cables and/or short-ranged wireless technology**
 - **Doesn't require a leased telecommunications line from an Internet Service Provider (ISP)**
- **Examples:**
 - **Home Network**
 - **Small Business or Office Network**

Wireless Local Area Network

(WLAN)

- **A LAN that's dependent on wireless connectivity or one that extends a traditional wired LAN to a wireless LAN**
- **Most home networks are WLANs**



Campus Area Network

(CAN)

- **A computer network of multiple interconnected LANs in a limited geographical area, such as a corporate business park, government agency, or university campus.**



Metropolitan Area Network

(MAN)

- **A computer network that interconnects users with computer resources in a city.**
- **Larger than a campus area network, but smaller than a wide area network**



Wide Area Network

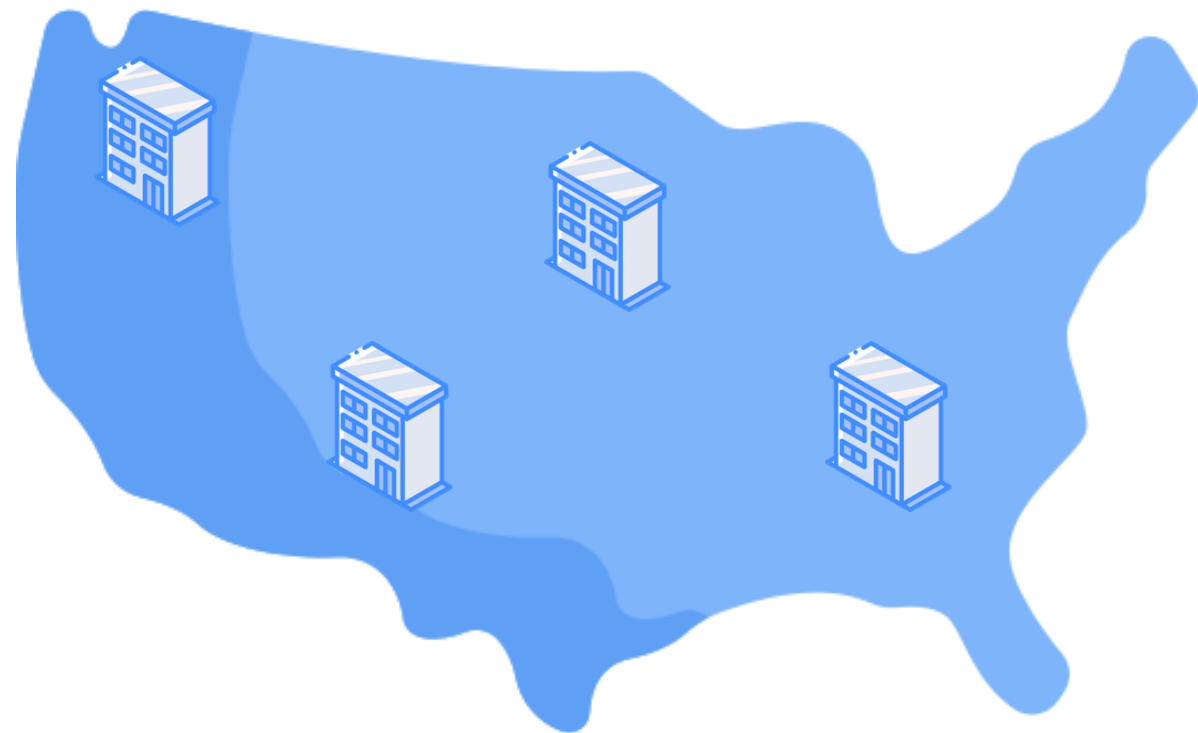


(WAN)

- **A computer network that extends over a large geographical distance, typically multiple cities and countries.**
- **WANs connect geographically distant LANs.**
- **Typically use leased telecommunications lines from ISPs.**

Examples:

- **The Internet**
- **Corporate Offices in Different States**



Types of Addresses



MAC Address

A MAC Address, often called a Physical Address, is a unique identifier assigned to a network interface controller (NIC) for communication within a network segment. The first three bytes are assigned by the IEEE to the manufacturing, the last three are unique numbers. Mac address is usually used to identify a device in an investigation

Example: 00-15-5D-D6-54-87

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::42ca:57b6:69c7:
IPv4 Address. . . . . : 192.168.1.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1%3
                             192.168.1.1
```

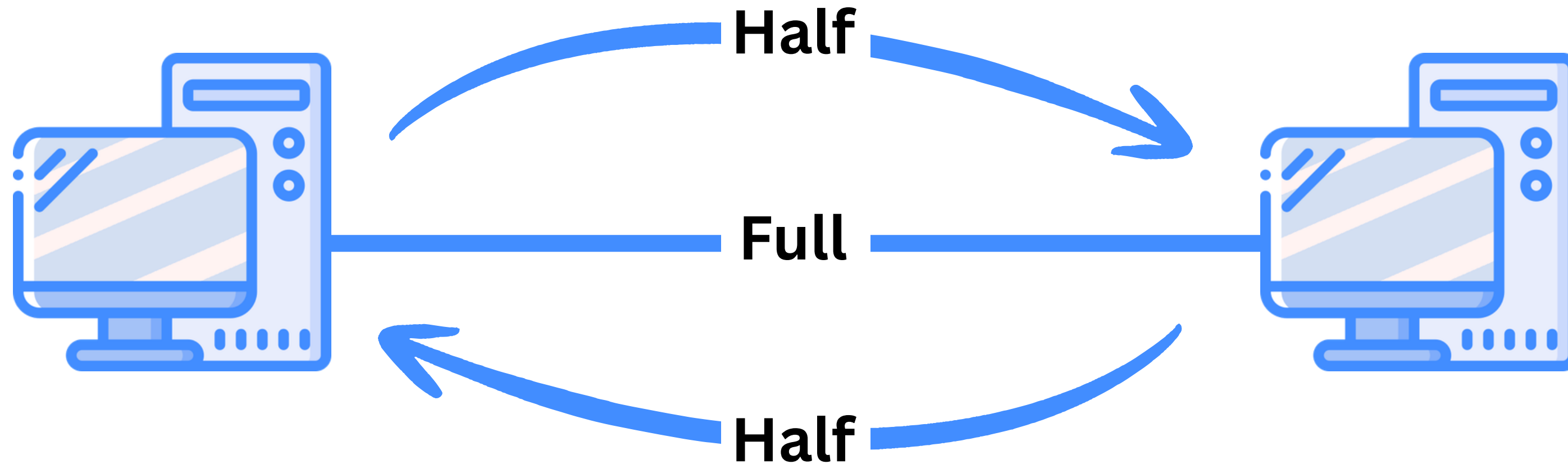
IP Address

A logical address, such as an IP address, is used to uniquely identify a device on a network. This address is assigned dynamically or statically, allowing devices to be located and communicate with each other across various networks, including the internet.

Examples:

- 1. 192.168.1.10**
- 2. 66.249.68.33**

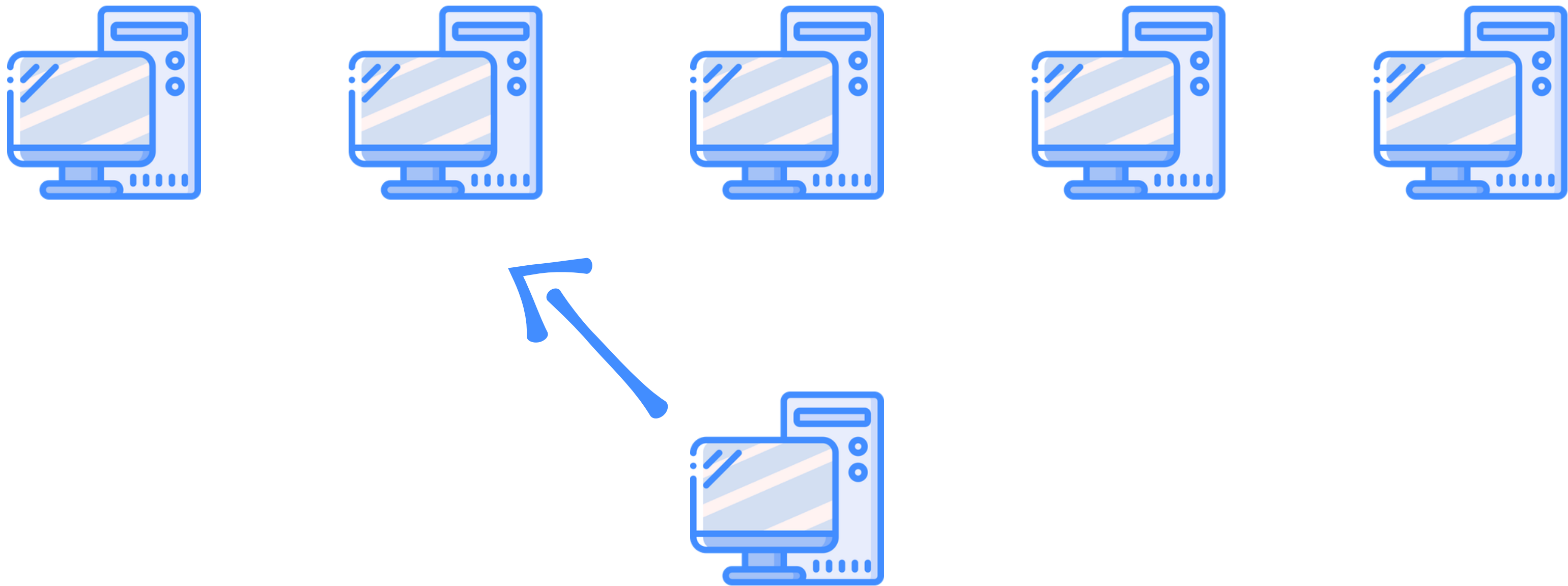
Half vs. Full Duplex Communication



- **Network communication will occur in either full or half duplex mode:**
 - **Half Duplex: Can send and receive data, but not at the same time**
 - **Full Duplex: Can send and receive data simultaneously**

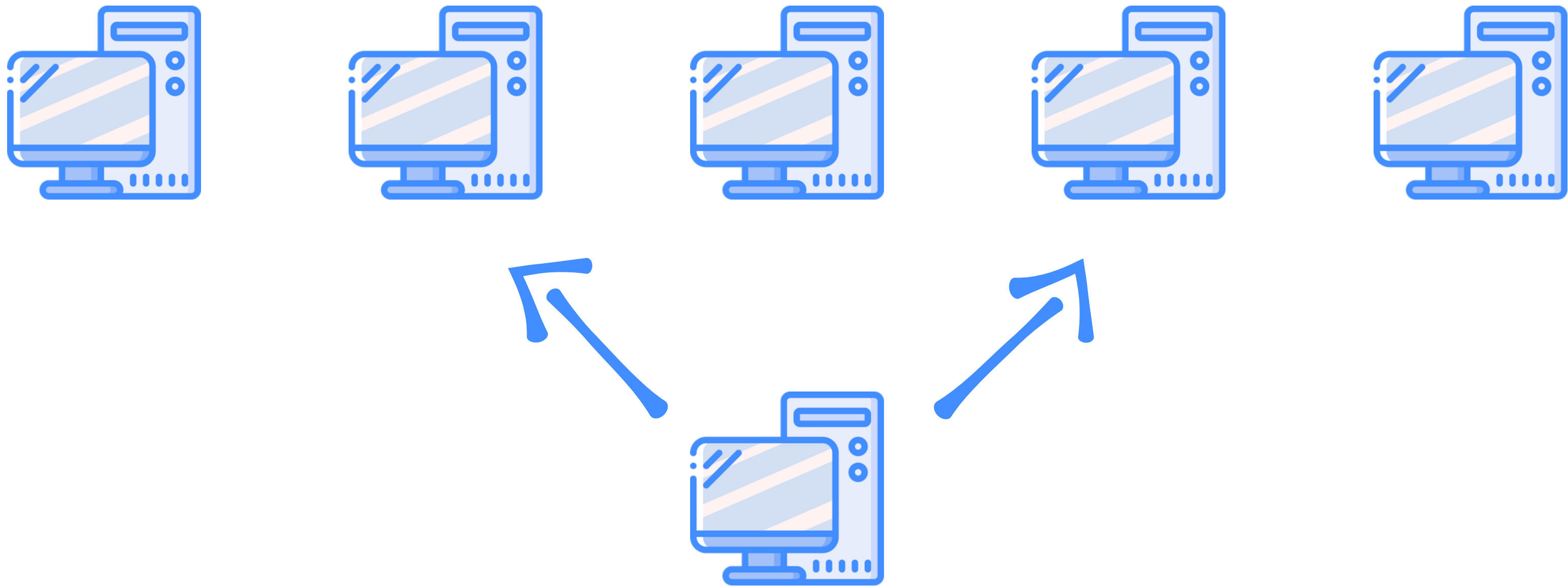
Network Transmission Types

UNICAST (One - to - One)



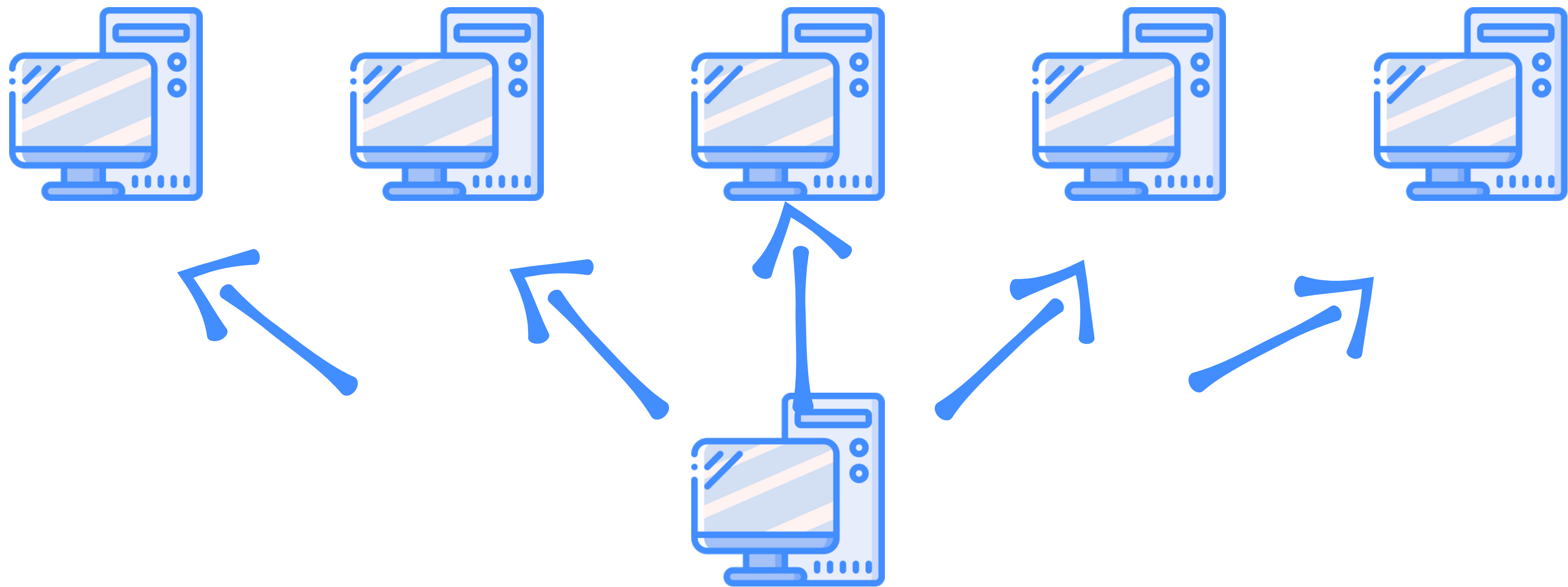
Network Transmission Types

MULTICAST (One - to - Many)

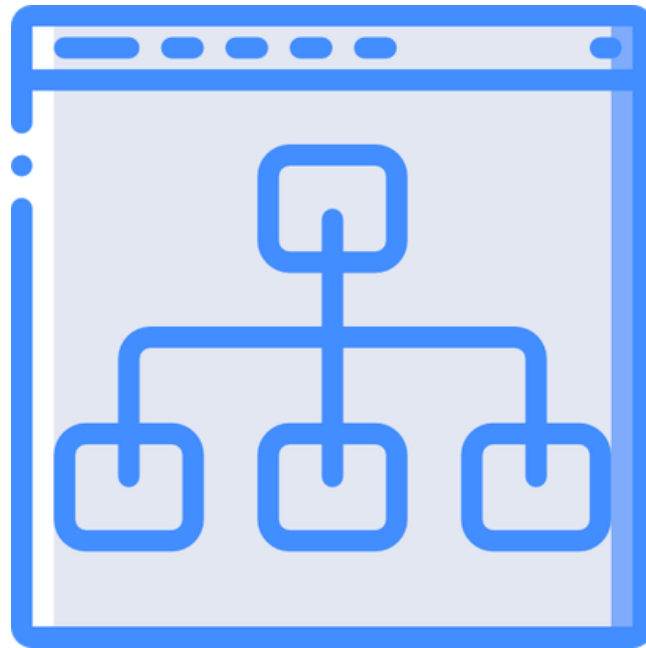


Network Transmission Types

BROADCAST (One - to - All)



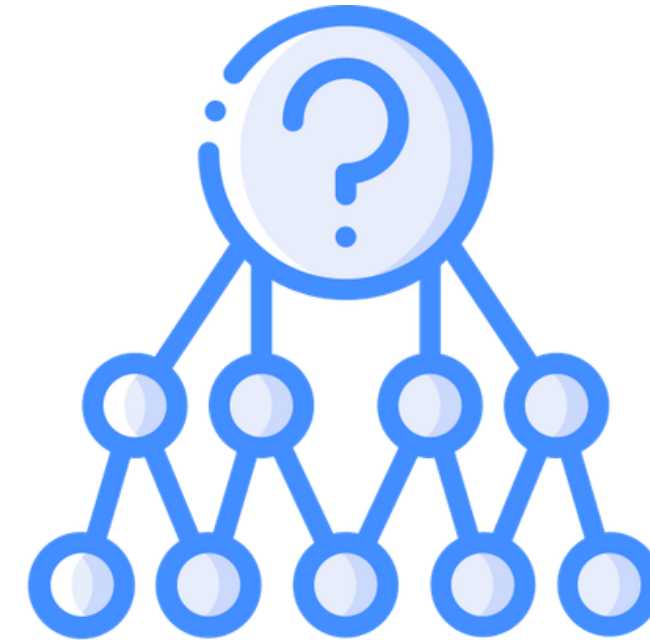
Physical vs Logical Diagram



Physical

A physical connection is the link between network devices.

Examples: Cables, Ethernet, fiber, or coaxial cables, radio waves (Wi-Fi, Bluetooth) or infrared signals



Logical

A logical connection is the **data being transported over a physical connection. Logical connection ensures that data arrives at the correct destination in an organized way using **protocols****